Bayes Theorem and Conditional Probability

Detailed Exam Notes (3–5 Pages)

1. Introduction to Probability

Probability is a measure of the chance that an event will occur.
The value of probability always lies between 0 and 1.

Formula:
$P(E)$ = Number of favourable outcomes / Total number of outcomes

Example:
When a die is rolled, the probability of getting 4 is:
$P(4) = 1/6$

2. Conditional Probability

Conditional probability is the probability of an event occurring when another event has already occurred.

Formula:
$P(A|B) = P(A \cap B) / P(B)$

Where:
• $P(A|B)$ = Probability of A given B
• $P(A \cap B)$ = Probability of both A and B occurring
• $P(B)$ = Probability of B

Example:
A card is selected from a deck. If it is known that the card is a face card,
find the probability that it is a king.

Face cards = 12
Kings = 4

P(King | Face Card) = 4/12 = 1/3

3. Bayes Theorem

Bayes Theorem helps us revise probabilities when new information becomes available.

Formula:
$P(A|B) = [P(B|A) \times P(A)] / P(B)$

Terms:
• $P(A)$ = Prior Probability
• $P(B|A)$ = Likelihood
• $P(B)$ = Evidence

• $P(A|B)$ = Posterior Probability

Applications:
• Medical diagnosis
• Machine Learning
• Spam email detection
• Risk analysis

Example:
Suppose 1% of people have a disease.
A test detects the disease correctly 99% of the time.
Find the probability that a person actually has the disease if the test result is positive.

Bayes theorem is used to update the probability after obtaining test results.

4. Multiplication Rule

The multiplication rule helps calculate the probability of two events occurring together.

$P(A \cap B) = P(A) \times P(B|A)$

Also,
$P(A \cap B) = P(B) \times P(A|B)$

Example:
A bag contains 3 red and 2 blue balls.
Two balls are drawn without replacement.

P(Red then Blue)
= P(Red) × P(Blue|Red)
= $(3/5) \times (2/4)$
= $3/10$

5. Independent Events

Two events are independent if occurrence of one event does not affect the other.

Formula:
$P(A \cap B) = P(A) \times P(B)$

For three independent events:
$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

Example:
Tossing a coin and rolling a die are independent events.

P(Head and 4)
= $1/2 \times 1/6$
= $1/12$

6. Complement Rule

The complement of an event means the event does not occur.

Formula:
$P(A') = 1 - P(A)$

Example:
If probability of rain tomorrow is 0.3,
then probability of no rain is:

$P(Rain') = 1 - 0.3$
$= 0.7$

7. Sample Space

Sample Space is the set of all possible outcomes.

Notation:
S = {All possible outcomes}

Example:
When a die is rolled:
$S = \{1, 2, 3, 4, 5, 6\}$

Number of outcomes = 6

8. Hypothesis Testing Concepts

A hypothesis is an assumption about a population parameter.

Two situations:
• Hypothesis is True
• Hypothesis is False

Decision options:
• Accept hypothesis
• Reject hypothesis

9. Type I Error and Type II Error

Type I Error (False Positive):
Rejecting a true hypothesis.

Example:
A healthy person is diagnosed as sick.

Type II Error (False Negative):
Accepting a false hypothesis.

Example:
A sick person is diagnosed as healthy.

Summary:

True Hypothesis + Reject = Type I Error
False Hypothesis + Accept = Type II Error

10. Important Exam Formulas

1. $P(A|B) = P(A \cap B)/P(B)$

2. $P(B|A) = P(A \cap B)/P(A)$

3. $P(A|B) = [P(B|A) \times P(A)]/P(B)$

4. $P(A \cap B) = P(A) \times P(B|A)$

5. $P(A \cap B) = P(B) \times P(A|B)$

6. $P(A \cap B) = P(A) \times P(B)$ (Independent Events)

7. $P(A') = 1 – P(A)$

8. $P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

Quick Revision Sheet

• Conditional Probability → Probability of A given B.
• Bayes Theorem → Updates probability using new evidence.
• Independent Events → One event does not affect another.
• Complement Rule → $P(A') = 1 - P(A)$.
• Type I Error → False Positive.
• Type II Error → False Negative.
• Sample Space → Set of all possible outcomes.